

Week 3: Retrieval-Augmented Generation (RAG)

Learning Topics

- Introduction to Retrieval-Augmented Generation (RAG)
- RAG Architecture
- Embeddings
- Embedding Models
- Vector Databases
- FAISS
- ChromaDB
- Document Loaders
- Text Splitters
- Chunking Strategies
- Similarity Search
- Retrievers
- Basic RAG Pipeline

Weekly Task

Build a **DStarix Internship Guide Chatbot** using **Retrieval-Augmented Generation (RAG)** that can answer user questions based on the provided internship guide document. The chatbot should retrieve relevant information from the document and generate accurate responses using any one of the following LLM APIs (**OpenAI, Gemini, Claude, or any free LLM**).

The application can be implemented as either a **Command-Line Interface (CLI)** or a **Web Interface**.

Expected Deliverables

- Complete source code

- GitHub repository
- README.md with setup instructions
- requirements.txt
- Environment variables configured using a .env file
- Internship Guide document used for the knowledge base
- Working RAG Chatbot capable of answering internship-related questions accurately

Submission

Submit the **GitHub repository link** before the assigned deadline. Ensure the repository includes all required files, documentation, and the Internship Guide document used for the RAG knowledge base.